

Introduction to Statistics - Homework #4

Exercise 1

An employee at a call center takes between 0.5 and 6.5 minutes to respond to a phone call, and this response time is assumed to be uniformly distributed between 0.5 and 6.5. If the response times of 60 independent calls are observed, approximate the probability that the average response time is at least 4 minutes.

Solution: Since X is uniformly distributed between 0.5 and 6.5, its pdf is $f(x) = \frac{1}{6}$ for $0.5 \leq x \leq 6.5$. Then,

$$E(X) = \int_{0.5}^{6.5} x \cdot \frac{1}{6} dx = \frac{1}{6} \left[\frac{x^2}{2} \right]_{0.5}^{6.5} = \frac{1}{12} (6.5^2 - 0.5^2) = \frac{42}{12} = 3.5,$$

$$E(X^2) = \int_{0.5}^{6.5} x^2 \cdot \frac{1}{6} dx = \frac{1}{6} \left[\frac{x^3}{3} \right]_{0.5}^{6.5} = \frac{1}{18} (6.5^3 - 0.5^3) = \frac{274.5}{18} = 15.25,$$

$$Var(X) = E(X^2) - (E(X))^2 = 15.25 - 3.5^2 = 15.25 - 12.25 = 3.$$

These agree with the standard formulas $E(X) = \frac{a+b}{2} = 3.5$ and $Var(X) = \frac{(b-a)^2}{12} = 3$ for the uniform distribution on $[a, b]$.

Thus,

$$E(\bar{X}) = 3.5, \quad Var(\bar{X}) = \frac{3}{60} = 0.05.$$

With a sample size of $n = 60$, which is sufficiently large, by the Central Limit Theorem we approximate

$$\bar{X} \sim N(3.5, 0.05).$$

Hence,

$$Z = \frac{\bar{X} - 3.5}{\sqrt{0.05}} \sim N(0, 1).$$

Therefore, the desired probability is

$$P(\bar{X} \geq 4) = P\left(\frac{\bar{X} - 3.5}{\sqrt{0.05}} \geq \frac{4 - 3.5}{\sqrt{0.05}}\right) = P(Z \geq 2.2361) \approx 0.0127.$$

Exercise 2

When Team A and Team B play a game in a sport, probabilities that Team A wins or loses are 0.6 and 0.4, respectively. Each outcome gives 3 points for a win and 0 points for a loss. Let $X_1, X_2, \dots, X_n$ denote the points earned by Team A in n independent games (assume independence across games). Answer the following questions.

- (1) Find the joint probability mass function of X_1 and X_2 .

Solution:

Since the games are independent, the joint probability distribution table is:

$x_1 \setminus x_2$	0	3
0	$0.4 \times 0.4 = 0.16$	$0.4 \times 0.6 = 0.24$
3	$0.6 \times 0.4 = 0.24$	$0.6 \times 0.6 = 0.36$

- (2) Let $\bar{X}_2$ be the average score over 2 games. Find the probability mass function of $\bar{X}_2$.

Solution:

$$f(x) = \begin{cases} 0.16, & x = 0, \\ 0.48, & x = 1.5, \\ 0.36, & x = 3, \\ 0, & \text{otherwise.} \end{cases}$$

- (3) Find the expectation $E[\bar{X}_2]$ and variance $Var(\bar{X}_2)$.

Solution:

First, we find the mean and variance of X_1 :

$$E[X_1] = 3 \times 0.6 + 0 \times 0.4 = 1.8,$$

$$E[X_1^2] = 3^2 \times 0.6 + 0^2 \times 0.4 = 5.4,$$

$$Var(X_1) = E[X_1^2] - (E[X_1])^2 = 5.4 - 1.8^2 = 5.4 - 3.24 = 2.16.$$

Therefore,

$$E[\bar{X}_2] = E[X_1] = 1.8, \quad Var(\bar{X}_2) = \frac{Var(X_1)}{2} = \frac{2.16}{2} = 1.08.$$

- (4) Suppose the two teams play 54 games. Approximate the probability that Team A's average score is at least 2.

Solution:

Since the sample size $n = 54$ is large, by the Central Limit Theorem:

$$\bar{X}_{54} \sim N\left(1.8, \frac{2.16}{54}\right).$$

Equivalently,

$$Z = \frac{\bar{X}_{54} - 1.8}{\sqrt{2.16/54}} \sim N(0, 1).$$

Therefore,

$$P(\bar{X}_{54} \geq 2) = P\left(\frac{\bar{X}_{54} - 1.8}{\sqrt{2.16/54}} \geq \frac{2 - 1.8}{\sqrt{2.16/54}}\right) = P(Z \geq 1) \approx 0.1587.$$

Exercise 3

The weight (in g) of a product follows $N(\mu, 3^2)$. Take a random sample of size $n = 16$ and let S^2 be the sample variance. Define

$$V = \frac{(n-1)S^2}{\sigma^2}.$$

Answer the following questions.

(1) Find the distribution of V .

Solution:

Since the population is normal,

$$V = \frac{(n-1)S^2}{\sigma^2} = \frac{15S^2}{9} \sim \chi^2(n-1) = \chi^2(15).$$

That is, V follows the chi-square distribution with 15 degrees of freedom.

(2) Find the probability $P(V < 25)$.

Solution:

From the chi-square table, $\chi_{0.05}^2(15) = 24.996 \approx 25$, so $P(V > 25) \approx 0.05$. Therefore,

$$P(V < 25) = 1 - P(V > 25) \approx 1 - 0.05 = 0.95.$$